

A Three-Stage Data Quality and Annotation Validation Framework for Structural Defect Detection

This document presents a structured three-stage framework for ensuring high-quality datasets of structural defects. The pipeline integrates automated image-quality filtering, human defect-visibility assessment, and manual annotation verification. The goal is to improve dataset reliability, reduce noise in training data, and ensure consistent, high-quality polygon annotations for deep learning-based defect detection systems.

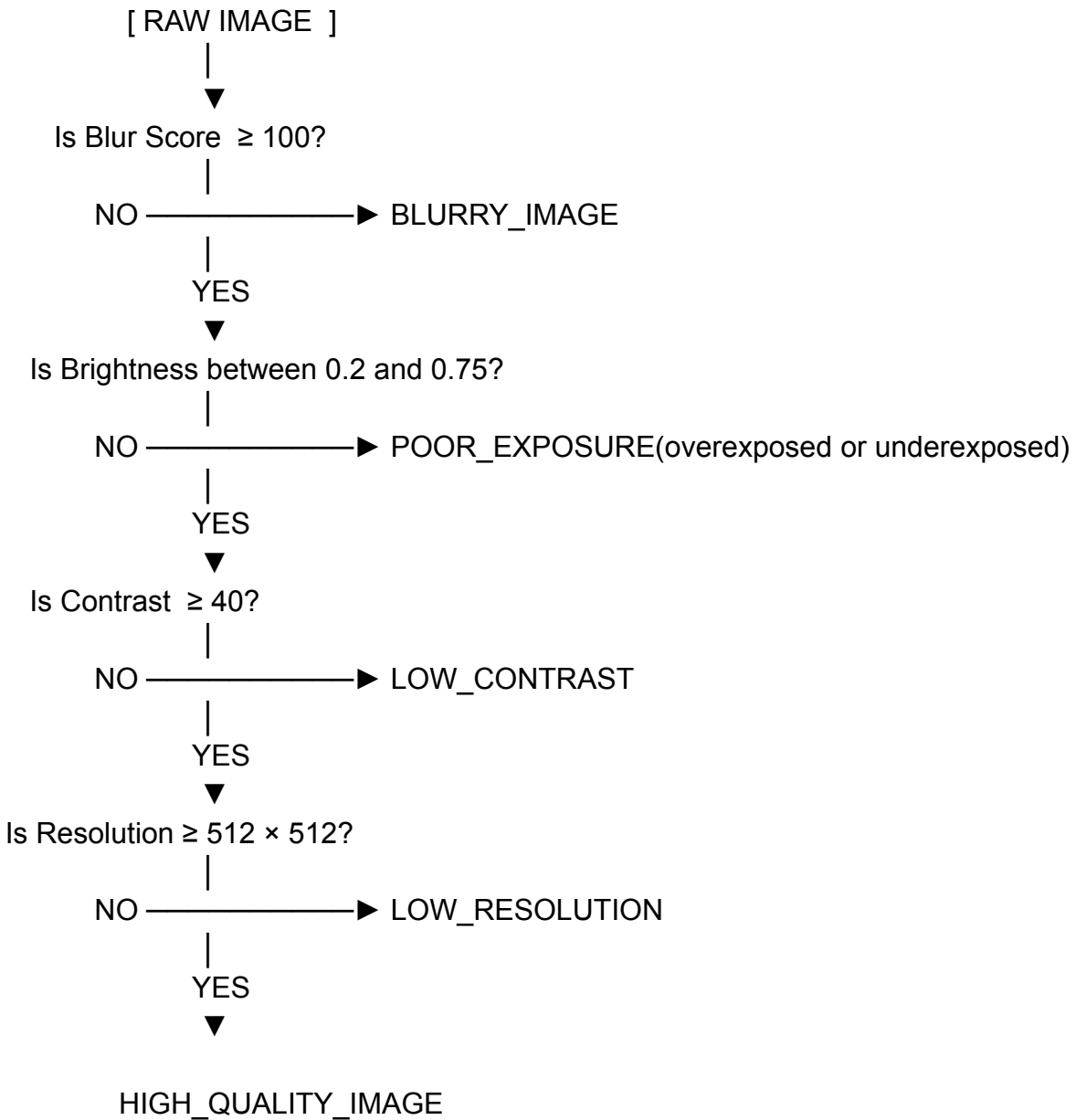
1. Introduction

Image-based structural inspection has become an important approach for automated defect detection in civil infrastructure. However, real-world datasets often contain images with low resolution, motion blur, poor illumination, and inconsistent annotations. These issues reduce model performance and introduce noise during training.

To address these challenges, a three-stage quality-control pipeline that systematically filters images and validates annotations before dataset use are proposed as follows:

2. Stage 1: Automated Image Quality Filtering

Objective: Remove technically unsuitable images before human inspection.

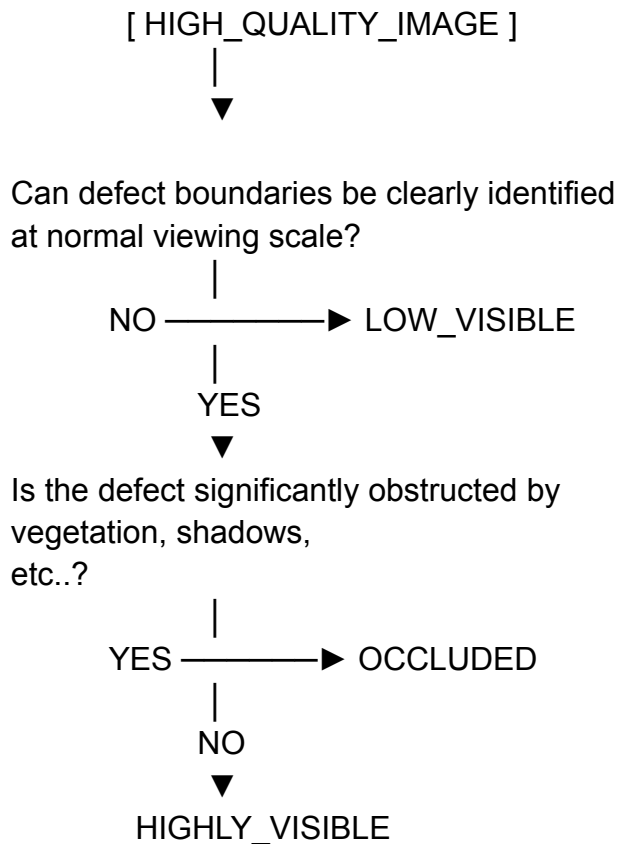


Stage 1 Output Tag name

Condition	Tag name
Low resolution	LOW_RESOLUTION
Motion blur	BLURRY_IMAGE
Over/under exposure	POOR_EXPOSURE
Low contrast	LOW_CONTRAST
Pass all criteria	HIGH_QUALITY_IMAGE

3. Stage 2: Human Defect Visibility Assessment

Objective: Evaluate whether defects are visually interpretable for annotation and model training.



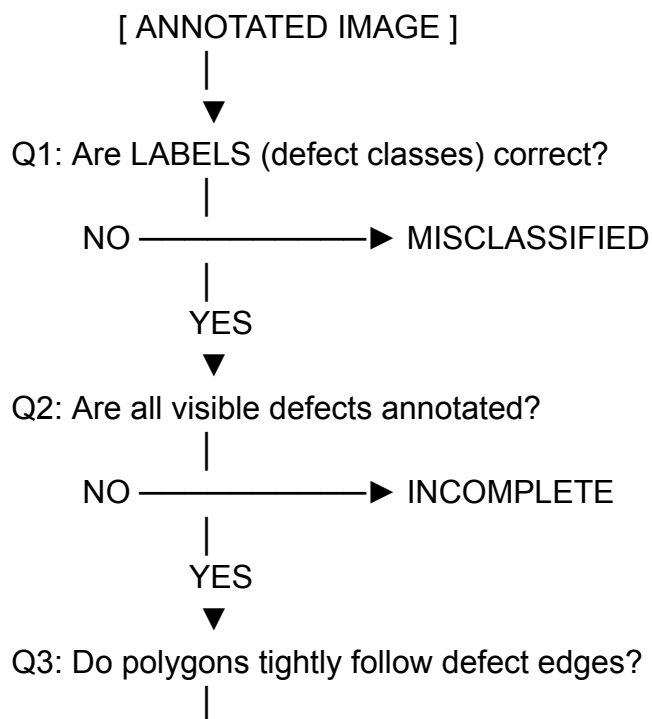
Stage 2 Output Tag name

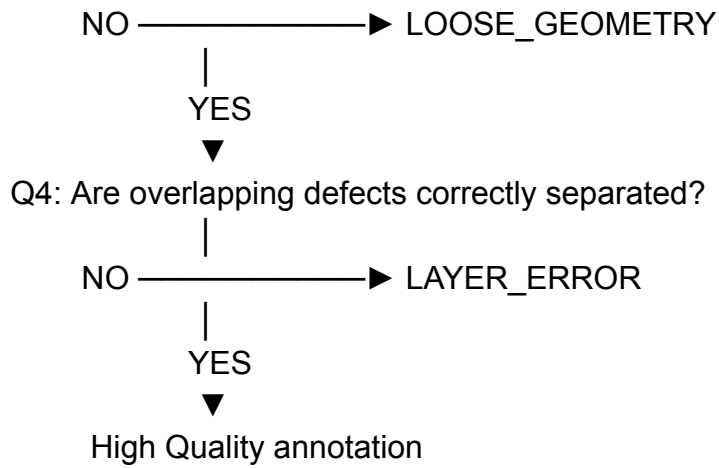
Condition	Tag name
Defect unclear	LOW_VISIBLE
Obstructed defect	OCCLUDED
Clearly visible defect	HIGHLY_VISIBLE

Only HIGHLY_VISIBLE samples proceed to annotation or model training.

4. Stage 3: Manual Annotation Validation (Polygon Review)

Objective: Ensure annotation correctness, completeness, and geometric consistency.





Stage 3 Output Classes

Condition	Tag name
Wrong class assignment	MISCLASSIFIED
Missing annotations	INCOMPLETE
Poor polygon accuracy	LOOSE_GEOMETRY
Incorrect layering	LAYER_ERROR
Fully correct annotation	High Quality Annotation

5. End-to-End Pipeline Summary

1. Automated image quality filtering based on objective visual metrics.
2. Human defect visibility assessment to ensure interpretability.
3. Manual annotation verification to guarantee labeling correctness.

Only samples that pass through all stages are retained for model training and evaluation, as well as further annotation.

